

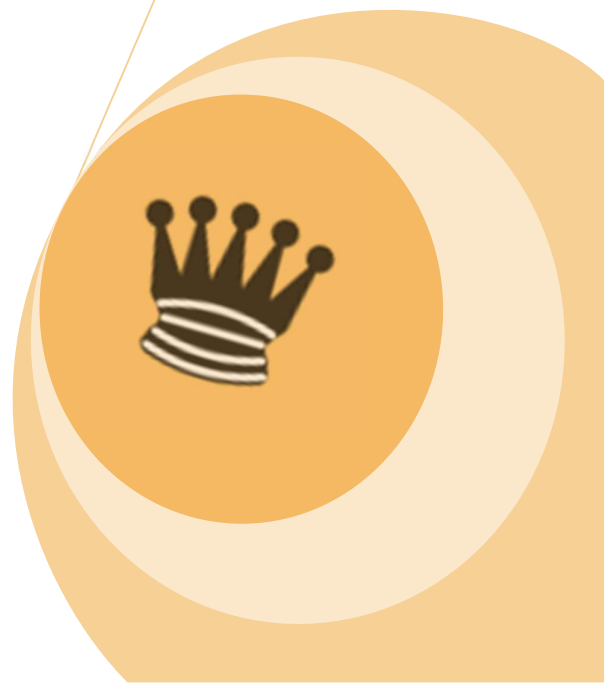


Chess Game & Artificial Intelligence

AQA GCE Computer Science NEA

Program to allow its users to play a game of chess, either with others or with a chess-playing AI I have created.

Alfie Jay Kunz
2021-2025



TRACKING AI PROGRESS

There are many methods of comparison to track how my AI has improved over time, such as search efficiency, ELO, online puzzle rating, etc. However, in this section I will be focussing on how long it takes each version of my AI to complete a search at a given depth, as I believe this method provides the most accurate and easily comparable scale of how my AI system has developed. To increase the accuracy of these tests, they will be performed on three sample chess positions: a typical opening position, a typical middlegame positions, and a typical endgame position, with white to move in all three cases.

To make the tests more apt for real-world comparisons, I will also calculate the centipawn loss for the AIs' moves. This is calculated by finding the difference between Stockfish's best move in the position, and Stockfish's evaluation of the AI's chosen move – I have chosen the Stockfish AI as the method of comparison as it is the undisputed greatest chess-playing entity in the world right now.

Each AI will search on the position two times in a row, and its lowest time of the two will be recorded. Its time will then be plotted onto a base 2 logarithmic graph (chosen as it best illustrates the differences between the AIs). For more information on these tests, along with extra comparative data, please visit the below link to access an online text document:

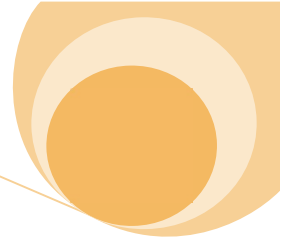
<https://drive.google.com/file/d/1YCkNUddRsQpQHxBxgHVB79Z01lelPjai>

WHERE TO GO FROM HERE...

A NOTE FROM ALFIE: 28/12/24

As you will be able to see, the improvements made in v9.0 have allowed it to absolutely dominate in the below tests. As a result, using this method for testing future versions will likely be un-fruitful. As I transition from upgrading the AI's ability to search, to the AI's ability to evaluate and order moves, we will likely see a large drop in search time, even though these features are making the AI stronger overall. Making this transition at this point in time seems wise, as the AI is now able to see *far enough* into the future to see tactics, but often overlooks basic techniques, and can be easily overwhelmed by an opponents attack. If I implemented these evaluation upgrades in, for example, v5.2, or even v8.3 to some extent, we would likely hinder the AI's ability to see far enough into the future to see basic tactics / checkmating patterns, possibly making the engine **weaker** overall.

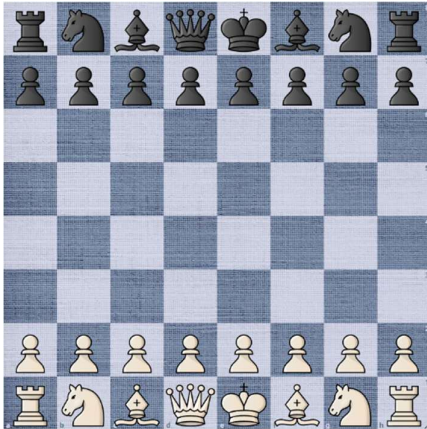
What these evaluation upgrades will be, and how I plan to compare & test these new versions against each other (and other bots, for that matter) will hopefully be thought about soon :) (although, with the addition of Remote Mode in v9.0, and hence being able to play against other bots on any chess interface, very easily, testing these AIs should be made much easier). But at the moment, I'm starting to see chess pieces in my dreams haha, so I'll leave that for future Alfie to figure out. With that being said, thanks so much to those that are reading this, and I hope that you win find / have found my project interesting. Until next time! :) -Alfie



(tests on next page)

OPENING TEST

The position used for the opening test will be the chess starting position, denoted in FEN notation with (rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR w KQkq -). This was chosen for the position's simplicity and greatest real-game comparison. Our AI will search on this position with a static depth of 5, and 7 (for the stronger AIs).



Sample Opening Position.

Best Move (Stockfish) = [e4, Nf3, d4]

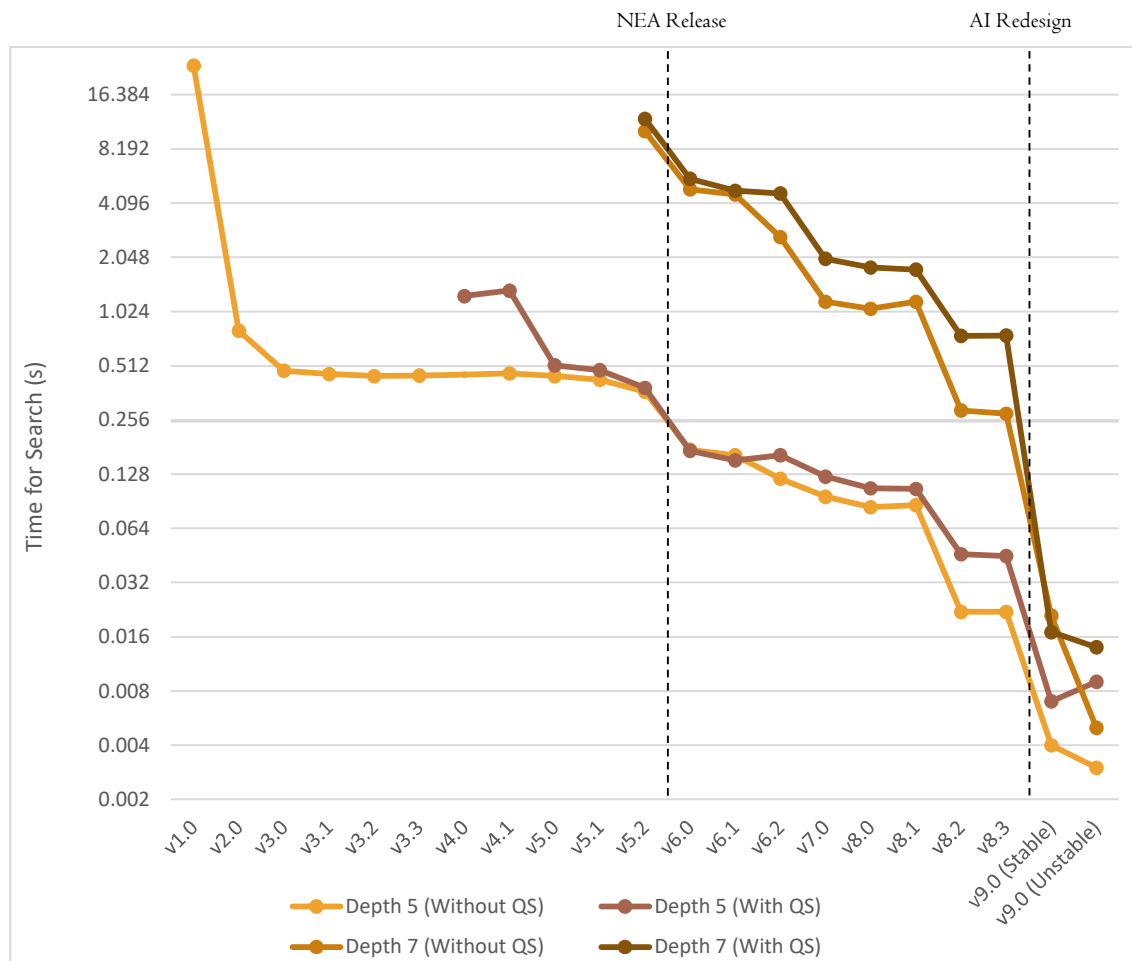
Current Results: (v9.0 – Unstable)

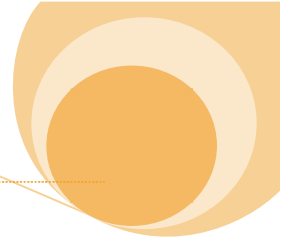
Depth 5 (without Quiescence): 1,267 positions searched in 0.003 seconds.
Move = a3 (centipawn loss = 30).

Depth 5 (with Quiescence): 1,495 positions searched in 0.009 seconds.
Move = a3 (centipawn loss = 30).

Depth 7 (without Quiescence): 1,931 positions searched in 0.005 seconds.
Move = a3 (centipawn loss = 30).

Depth 7 (with Quiescence): 2,468 positions searched in 0.014 seconds.
Move = a3 (centipawn loss = 30).





MIDDLEGAME TEST

The middlegame position is this complex position, denoted in FEN notation with (r3k2r/p1ppqpb1/bn2pnp1/3PN3/1p2P3/2N2Q1p/PPPBPPPP/R3K2R w KQkq -). This position was chosen for its complexity, and for having many possible moves from a wide range of evaluations. Our AI will search on this position with a static depth of 4, and 5 (for the stronger AIs).



Sample Middlegame Position.

Best Move (Stockfish) = [Bxa6, dxe6]

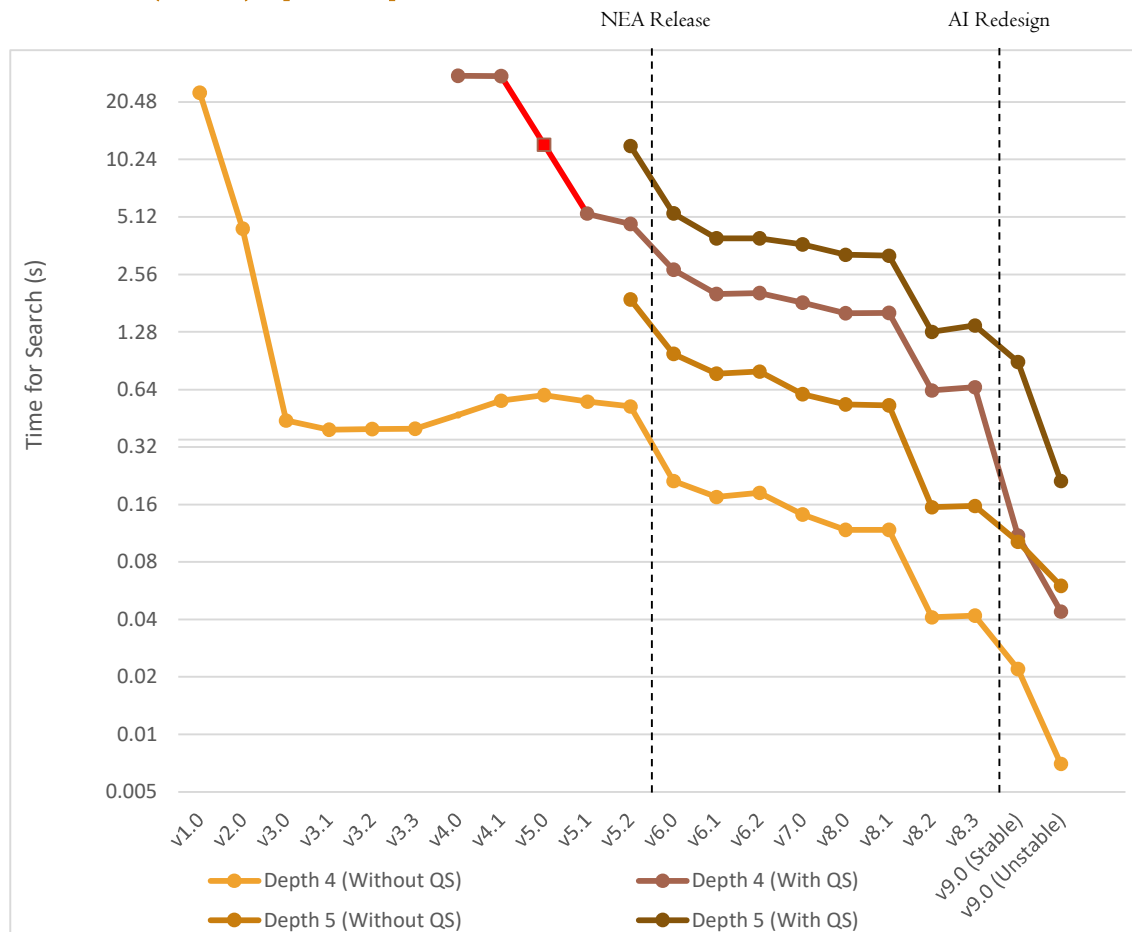
Current Results: (v9.0 – Unstable)

Depth 4 (without Quiescence): 3,063 positions searched in 0.007 seconds.
Move = Bxa6 (centipawn loss = 0).

Depth 4 (with Quiescence): 6,954 positions searched in 0.044 seconds.
Move = Bxa6 (centipawn loss = 0).

Depth 5 (without Quiescence): 10,717 positions searched in 0.060 seconds. Move = Bxa6 (centipawn loss = 0).

Depth 5 (with Quiescence): 27,755 positions searched in 0.212 seconds.
Move = Bxa6 (centipawn loss = 0).



ENDGAME TEST

The final position is a medium-complex endgame, denoted in FEN notation with (2r3k1/8/6p1/ppP3Kp/4RP1P/6P1/P7/8 w -). This position was chosen due to the limited number of moves that are winning for white, as well as it striking a good balance of complexity. Our AI will search on this position with a static depth of 6.



Sample Endgame Position.

Best Move (Stockfish) = [Kxg6]

Current Results: (v9.0 – Unstable)

Depth 6 (without Quiescence): 4,277 positions searched in 0.011 seconds.

Move = Kxg6 (centipawn loss = 0).

Depth 6 (with Quiescence): 6,117 positions searched in 0.029 seconds.

Move = Kxg6 (centipawn loss = 0).

Depth 8 (without Quiescence): 22,902 positions searched in 0.060 seconds. Move = Kxg6 (centipawn loss = 0).

Depth 8 (with Quiescence): 61,274 positions searched in 0.286 seconds.

Move = Kxg6 (centipawn loss = 0).

